Do Now · retrieval

Lesson 054 · Hash Tables and Dictionary<TKey,TValue>

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Which data structure uses FIFO in BFS?

Q2

What does a dictionary map?

Q3

What is a collision in hashing?

Q4

A hash table has slots 0–6, all empty. Insert keys 9, 16 and 23 in that order using h(k)=k MOD 7 and linear probing to the next slot. Determine the final slot occupied by 23.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Discover — A hash function maps keys to hash values that guide placement/lookups. A collision occurs if distinct keys map to the same location/bucket.

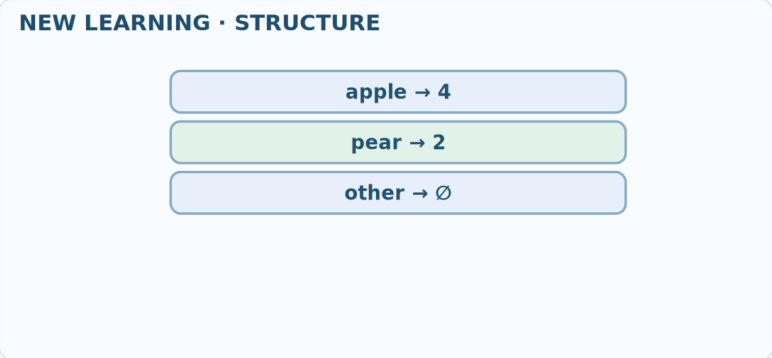
Core idea 2

Trace — A .NET Dictionary<TKey,TValue> handles collision management internally; do not equate its actual buckets with simplistic modulo examples.

Core idea 3

Evaluate — To read a missing key safely, use TryGetValue. Hash-table operations are often expected O(1) but poor collisions can degrade performance.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

hash function — Maps keys to hash values.

collision — Different keys share a target bucket.

bucket — Hash table storage location.

load factor — Filled entries relative to capacity.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

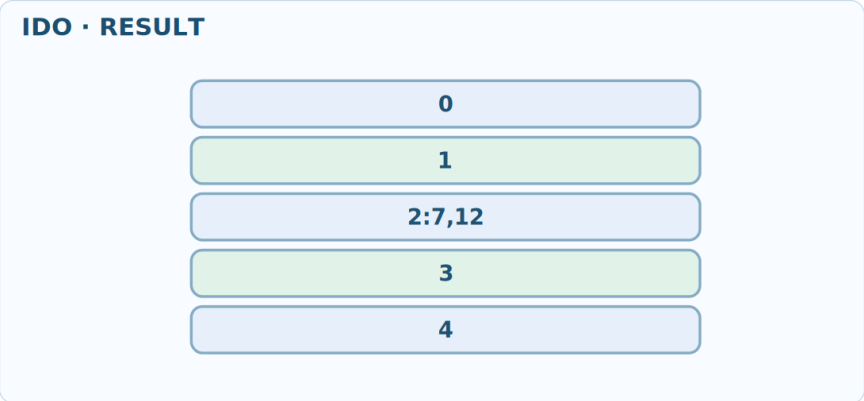
Problem

Apply a toy hash h(k)=k mod 5 to keys 7 and 12. What happens?

Follow the worked method

1. Explain the trace — Both keys collide in bucket 2; a table must resolve collision by chaining or probing.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Explain the trace
No collision: key 8 maps bucket 3, key11 maps bucket1.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

With toy h(k)=k mod5, find buckets for keys 8,13,19. [6]

Question 2

Write C# Dictionary<string,int> code storing marks Ada80 and Ben65, then safely test whether Cleo exists. [6]

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Compare chaining and linear probing for toy hash collisions, and state two performance concerns. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Hash tables can never have collisions when keys are different.

Use key vocabulary from memory

1. Explain hash function in your own words.

2. Explain collision in your own words.

3. Explain bucket in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.